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Research on low recoil launch technology of small UAV with small arms mounted on it based on floating principle

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Abstract: The small unmanned aerial vehicles (SUAV) which is mounted with small arms have both the accurate firing ability of small arms, and the mobility, flexibility and concealment of SUAV. In order to reduce the influence of recoil on SUAV, we put forward a low recoil launch method based on floating principle. Take a certain type of rifle as a study object, we have established a virtual prototype model. Through simulation and firing test, the recoil reduction effect is verified. This method provides a feasible and effective technical way for small unmanned platform to mount weapons in service.

1. Introduction

Small unmanned aerial vehicles (SUAV) have the characteristics of good concealment, strong mobility. The appearance of SUAV has brought significant changes to the military fighting style.

Integrating information, detection and strike is the development direction of SUAV in the future. After weaponization, SUAV can carry out traditional reconnaissance missions, make timely response to the battlefield situation, provide effective firepower striking, and its combat effectiveness will be greatly improved.

Small arms are the weapons, which are equipped most widely. The SUAV mounted with small arms have both the accurate firing ability of small arms, and the mobility, flexibility as well as concealment of small unmanned aircraft. It is not only applicable to special operations, urban warfare and anti-terrorism operations, but also can be used to anti-UAV on the information battlefield in future.

But small arms will produce greater recoil in firing, especially automatic weapons. The effects of recoil keep in the process of automatic firing. This force will not only affect the firing accuracy, but also have a great impact on the flight stability of SUAV, even can cause flip or fall. Small arms developed only for SUAV, loses the characteristics of generality and universality of small arms. So, developing the technology of SUAV mounted with small arms firing of low recoil is very important for integrating information, detection and strike of SUAV.

Based on the floating principle, we put forward a low recoil launch method for SUAV. A certain type of rifle as the study object, a theoretical model and a virtual prototype model are established, Through simulation and firing test, the recoil reduction effect is verified. This method provides a feasible and effective technical way for small unmanned platform to mount weapons in service.

2. Theoretical model

The firing way of automatic weapons in the process of recoil is called floating technology. When the weapon is firing, the forward energy of the floating body will partially offset the backward energy of



gunpowder and gas, thus the recoil force of shooting is reduced. Based on the floating principle, we put forward a low recoil launch method for small unmanned platform, which uses the weapon as a recoil body and gives it a certain speed before shooting in order to offset the recoil energy of the weapon, so as to reduce the impact of recoil on the flight stability of SUAV. The theoretical model is shown in the Figure 1.

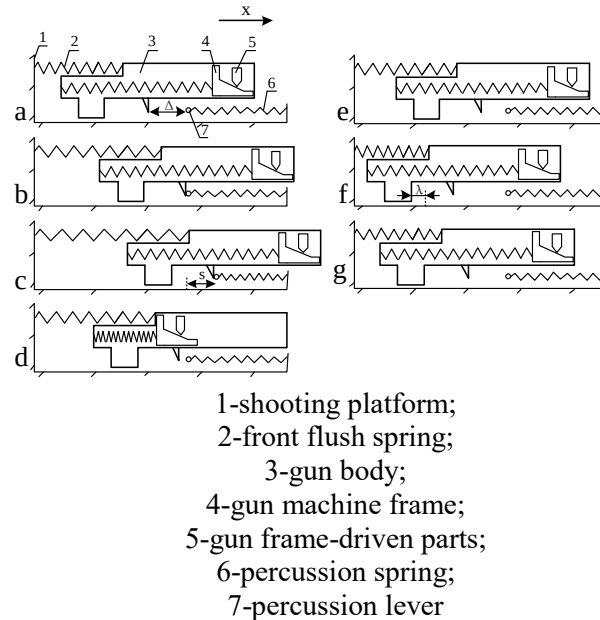


Figure 1. State diagram of the shooting platform

It works as follows:

During firing, rifle moves forward a distance Δ under the action of spring, then the trigger hits the cocking lever, when the pressure is greater than the trigger force, the hammer is released, the hammer hits the firing pin, and the primer is fired. After firing, the rifle stops moving forward, and begins to move back under the recoil force. At the same time, the weapon completes the actions, such as unlocking, pulling the shell, throwing the shell, pushing the shell and locking. When the rifle is rear-seated, and then spring force make the rifle to the starts position, a shooting cycle completed.

3. Establishment visual prototype

Virtual prototype technology can replace the physical model of the product for testing and improvement, which can not only shorten the time of research and development, but also reduce its expenditure. Therefore, we use ADAMS multi-body dynamics simulation software to build a virtual prototype of rifle shooting based on floating principle.

3.1. The basic assumptions

Movement of various components in the process of rifle shooting is complex, so it is impossible to take all the possible factors into account in the building virtual prototype. According to the purpose of this paper, we ignore the secondary factors and make the rifle virtual prototype, which does not affect rifle's normal shoot.

- Ignore the factors that have little effects on the motion of automat;
- Do not consider the resistance of the bullet to the barrel;
- Gunpowder gas exerts loads on the corresponding parts as an external force directly.

3.2. Visual prototype establishment

The model of visual prototype is shown as Figure 2:



Figure 2. The model of visual prototype

3.3. Load analysis

The forces that rifle bears in the process of shooting basically are resultant force of guns, air chamber pressure, air chamber pressure reaction force, all sorts of spring force.

3.3.1. Resultant force of guns

Resultant force of guns is the force of gunpowder gas acting on the muzzle, which can be divided into internal ballistic period and after-effect period. The resultant force can be got by converting the average chamber pressure to the chamber bottom pressure.

3.3.2. Pressure of gas-operated chamber

The chamber pressure can be calculated by Braven's empirical formula:

$$p_s = p_d' e^{\frac{a(t-t')}{b}} (1 - e^{\frac{a(t-t')}{b}}) (t' \leq t \leq t'') \quad (1)$$

In the formula, p_s pressure of gas-operated chamber, p_d' average pressures in the chamber when the bullet passes the gas-operated hole, b time parameters related to pressure of the bore, a parameters relating to structure size of gas-operated device, t time, t' the time of the bullet from start to gas hole, t'' the time of end of aftereffect period.

3.3.3. Extraction Resistance

Extraction resistance can be calculated by approximate calculation formula.

$$F_\phi = \pi \left\{ f_0 l_k \left[p d_1 + 2 E_1 \delta \left(\Delta - \frac{2x\alpha}{d_{pj}} \right) \right] - \frac{1}{4} d_1^2 p \right\} \quad (2)$$

In the formula, F_ϕ extraction resistance, f_0 friction coefficient between shell and chamber, l_k the length of shell, p pressure of propellant gas in cartridge case, d_1 diameter of cartridge case, E_1 inertial modulus of shell material, δ thickness of cartridge shell, Δ relative compaction between shell outer surface and shell wall, d_{pj} average diameter of shell, α semi-taper angle of conical part of shell, x backward travel of cartridge case.

3.4. Determination of Key Parameters

After the virtual prototype is built, it is necessary to optimize its key parameters so as to optimize the key parameters. Because the parameters of rifle can not be adjusted, so through sensitivity analysis, we determine the stiffness of the forward spring $k_q(N \cdot m^{-1})$, forward spring preload $F_{q0}(N)$, and forward distance $\Delta(mm)$ as optimization variables.

Table 1. Design variable value range

Variable	Initial value	Lower limit	Upper limit
$k_q/(N \cdot m^{-1})$	200	100	500
F_{q0}/N	90	10	100
Δ/mm	7	0	50

Using multi-island genetic algorithm, based on Isight software, a multi-disciplinary optimization platform, the dynamic simulation software Adams is invoked in batch mode to optimize the calculation.

Table 2. Design variables and objective function values before and after optimization

Variable	$k_q/(N \cdot m^{-1})$	F_{q0}/N	Δ/mm	F_{max}/N
Initial values	200	90	7	94.6
Optimization results	350	60.1	28.2	65.6

4. Simulation analysis

The dynamic simulation of the virtual prototype is carried out, and the velocity curve of the gun body is obtained as shown in Figure 3, and the recoil force curve of the shooting platform is shown in Figure 4.

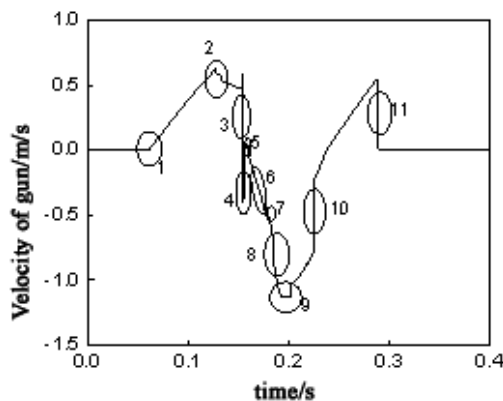


Figure 3. Velocity change curve of gun body

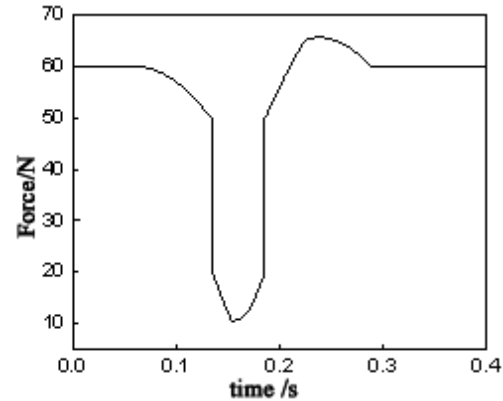


Figure 4. Recoil change curve of design platform

Mark 1 indicates that the rifle starts to move forward under the action of the forward spring; after marking 2 triggers impact the trigger rod, they collide with the casing under the action of the trigger spring, so there are two sudden changes in the speed of the rifle body; Mark 3 indicates that the combined force of the rifle chamber begins to work; Mark 4 indicates that the speed of the rifle body suddenly decreases, because the pressure reaction force of the gas-operated chamber is greater than the pressure in the rifle after the action of gunpowder gas. Mark 5 indicates that the rifle has been unlocked; Mark 6 is caused by shell throwing resistance; Mark 7 indicates that the hammer driven by the frame of the rifle collides with the buffer and begins to compress the buffer spring; Mark 8 indicates that the buffer collides with the butt driven by the frame and the hammer, and the frame of the rifle starts to recoil in place; Mark 9 marks that the rifle recoils and starts to recoil, and then strikes. The collision between hammer and single blocking iron results in sudden change of rifle speed; the collision between frame and casing occurs in location 10; the rifle restores in location 11, and the whole shooting is completed.

In order to compare the recoil reduction effect of the shooting platform, the maximum recoil force of the shooting platform during the shooting is compared with the maximum of the recoil force of the shoulder-proof shooting obtained from the experiment as shown in the table 3. The maximum recoil force of the shooting platform is only 16.2% of the maximum shoulder force, which shows that the recoil reduction effect of the low recoil shooting platform based on floating principle is very obvious.

Table 3. Comparison of the maximum value of front spring force and shoulder force

	Recoil force of the platform	Shoulder force
f_{max}/N	65.6	404.9

5. Firing test

In order to test the recoil reduction effect of the shooting platform, an experiments is designed to verify it. Firing test is shown as Figure 5. The velocity curve of rifle body during simulation and test is shown in Figure 6. The variation curve of recoil force of shooting platform is shown in Figure 7.



Figure 5. Firing test

It can be seen that the simulation data and the test data keep good consistency. The experimental curve of rifle speed in Figure 6 is not smooth because of the vibration of the shooting platform and the energy loss of the front spring. The influence of these factors can be ignored. There is a certain deviation between the simulation curve and the test curve of the recoil force of the shooting platform in Figure 7. This is because the friction force of the shooting platform is not uniform during the test. The friction force has little influence on the forward spring force, so it can be ignored.

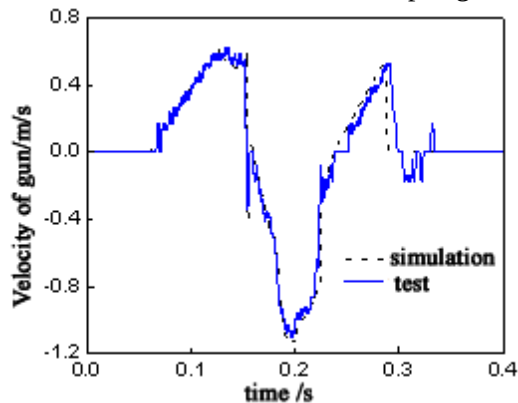


Figure 6. Comparison of gun body velocity simulation and test data

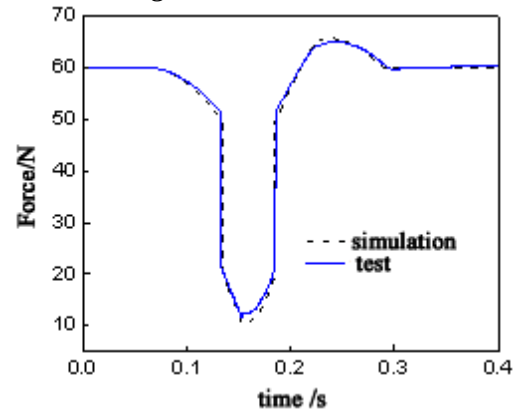


Figure 7. Comparison of recoil simulation and test data of design platform

The simulation data of several representative points of rifle are compared with the data in the firing test as shown in Table 4. v_1 the maximum forward speed of rifle, v_2 the maximum recoil speed of rifle, v_3 the maximum recoil speed of rifle, f_{max} the maximum recoil force of shooting platform. The error range between simulation data and test data is less than 2%, which meets the needs of engineering calculation, the result of virtual prototype is credible. Therefore, it can be concluded that the rifle shooting platform, based on floating principle, has a good recoil reduction effect.

Table 4. Comparison between simulation data and test data

	Simulation data	Test data	Error
$v_1/(m/s)$	0.63	0.62	1.6%
$v_1/(m/s)$	1.12	1.10	1.8%
$v_3/(m/s)$	0.53	0.52	1.9%
f_{max}/N	65.6	64.8	1.2%

6. Conclusion

In this paper, based on floating principle, a method of reducing recoil of the firing platform for the whole front impact firing is proposed, and a theoretical model is established. The virtual prototype model of rifle firing, based on the principle of whole rifle front impact firing, is established and simulated by adapting virtual prototype technology. The results show that the recoil force of rifle shooting decreases to 65.6N, which is only 16.2% of that of shoulder-to-shoulder shooting. The effect of recoil reduction is

very obvious. On this basis, firing test was carried out to verify the credibility of the simulation results. It is further illustrated that the rifle shooting platform based on the principle of whole-gun front-impact firing can effectively reduce the recoil force effect, which provides a feasible and effective technical way for the small unmanned platform to mount weapons in service.

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